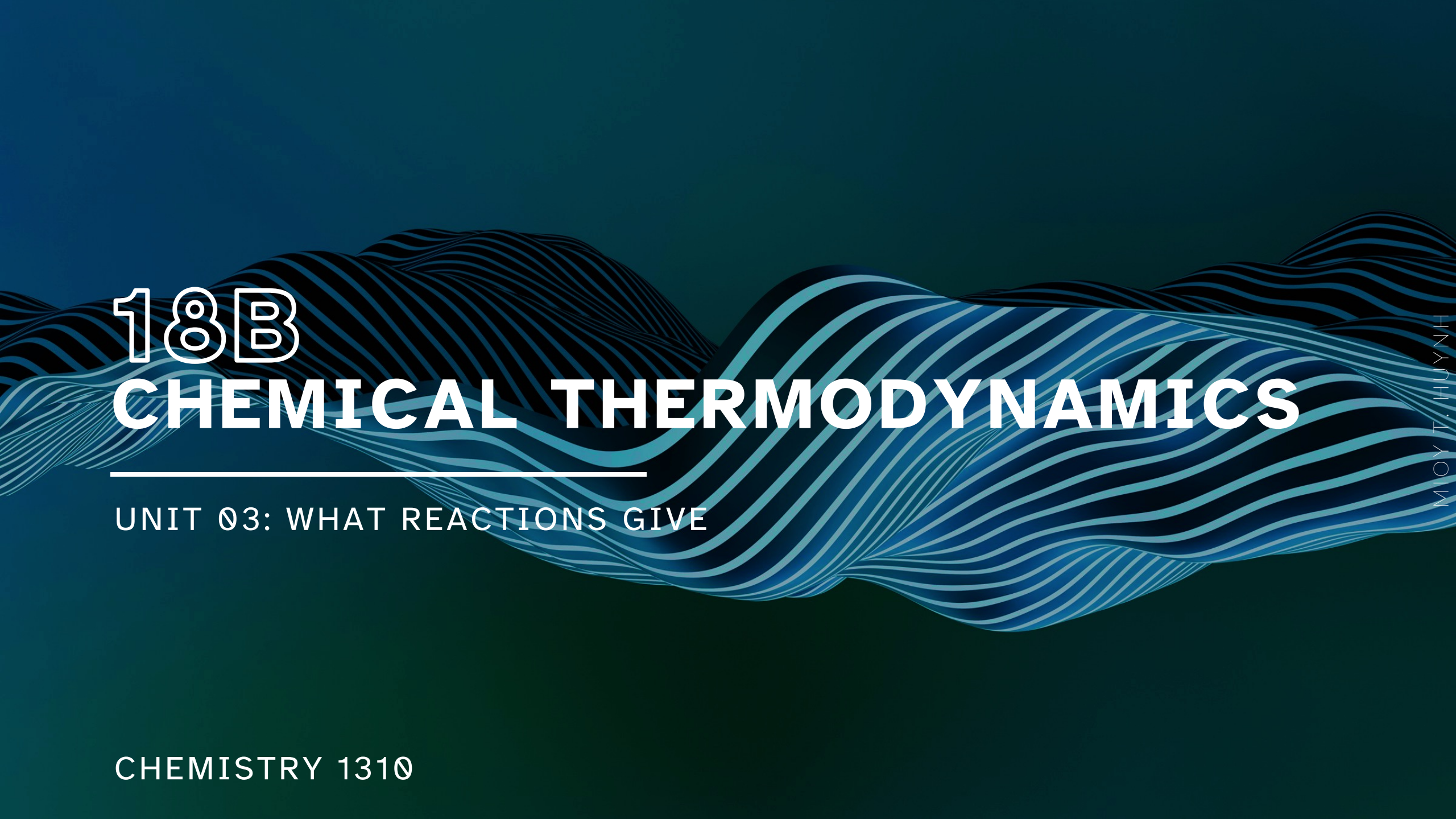




18B

CHEMICAL THERMODYNAMICS

UNIT 03: WHAT REACTIONS GIVE

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 18

1. State the second law of thermodynamics.
2. Define entropy (S) and identify the sign of ΔS from chemical equations or changes of state.
3. State the third law of thermodynamics.
4. Calculate ΔS_{rxn} from molar entropies.
5. Compare and contrast ΔS_{univ} , ΔS_{sys} , and ΔS_{surr} .
6. Define Gibbs free energy (G) and describe how the signs of ΔH and ΔS affect the sign of ΔG .
7. Relate the sign of ΔG to the favorability/spontaneity of a reaction.
8. Determine the equilibrium temperature of a reaction.
9. Apply the relationship between ΔG° and the equilibrium constant (K).

Entropy and Temperature

18.3

Second Law of Thermodynamics

- A *spontaneous process* is one that results in an overall *increase* in the entropy (disorder) of the universe.
- Since $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$, this means that

ΔS_{univ}	ΔS_{sys}	ΔS_{surr}
Spontaneous		
Spontaneous		
Spontaneous		
Nonspontaneous		

Gibbs Free Energy

18.4

Unfortunately, the Second Law is practically useless because it is often immeasurable.

So, how can the spontaneity of a process/reaction be predicted?

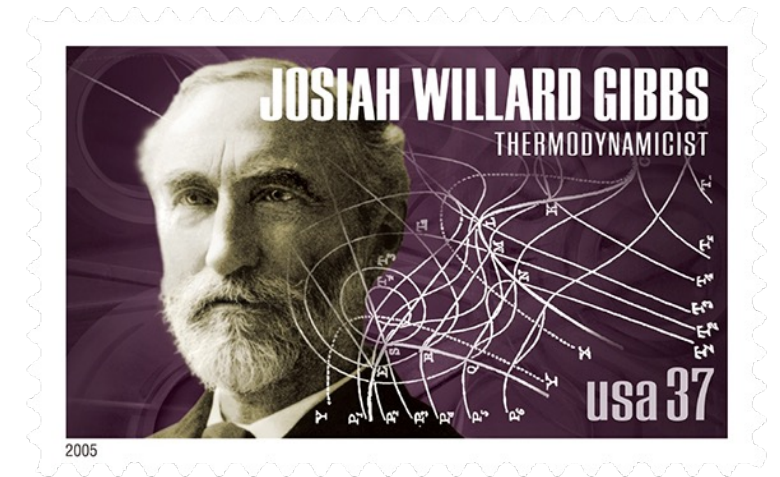
- *Exothermic processes ($\Delta H < 0$) are more likely to be spontaneous* than endothermic ($\Delta H > 0$) processes.
- Processes where there is an *increase in entropy ($\Delta S > 0$) are more likely to be spontaneous* than processes with decreases in entropy ($\Delta S < 0$).

Gibbs Free Energy

18.4

Gibbs Free Energy (G)

- A mathematical model to predict reaction spontaneity in terms of ΔH_{rxn} , ΔS_{rxn} , and T (temperature).
- This new state function is defined as



- The sign of ΔG_{rxn} is opposite the sign of ΔS_{univ} .
- Most importantly, under constant pressure conditions (i.e., benchtop reactions):

Recall the term “driving force” from earlier

Chemistry in Context

In Chapter 4, we said that reactions occur *spontaneously* due to a combination of changes in heat energy (ΔH_{rxn}) and randomness (ΔS_{rxn}).

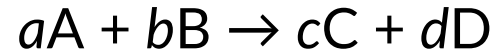
We also noted that certain *patterns* of reactants and products tend to be associated with *spontaneous reactions* and classified these patterns as *driving forces*.

- | | |
|---|---|
| 1. Formation of a precipitate | → Double-replacement reactions |
| 2. Formation of water or neutralization | → Acid-base reaction |
| 3. Oxidation-reduction | → Combustion, synthesis, decomposition, and single-replacements reactions |

But why the term driving force?

Gibbs Free Energy

18.4



Calculating changes in Gibbs free energy ($\Delta G_{\text{rxn}}^{\circ}$) from tabulated $\Delta G_{\text{f}}^{\circ}$ values.

Calculating changes in enthalpy ($\Delta H_{\text{rxn}}^{\circ}$) from tabulated $\Delta H_{\text{f}}^{\circ}$ values.

Calculating changes in entropy ($\Delta S_{\text{rxn}}^{\circ}$) from tabulated S° values.

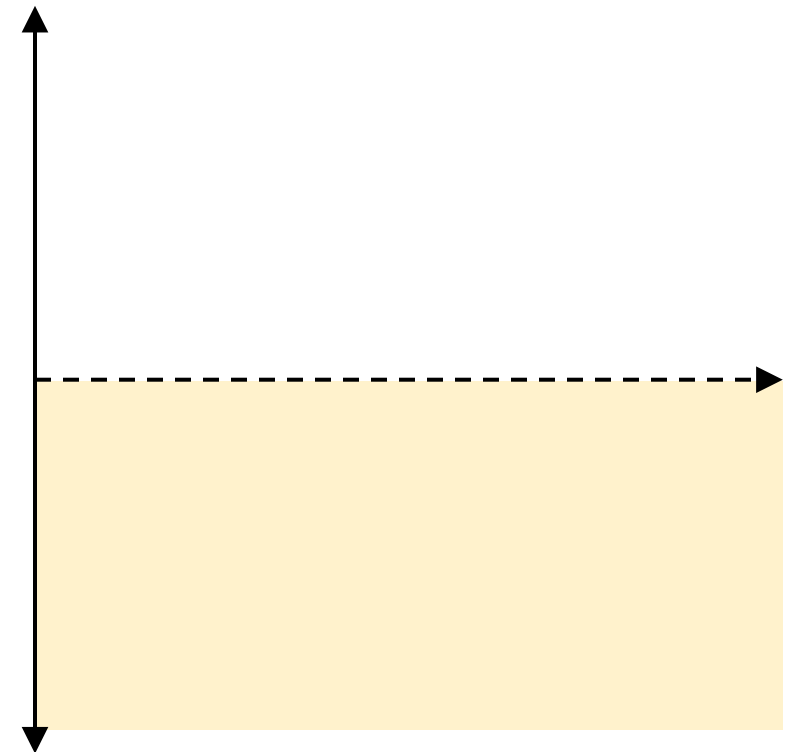
Free-Energy Changes and Temperature

18.5

For some processes, the value of ΔG_{rxn} is *temperature-dependent*.

So, these processes exhibit *temperature-dependent spontaneity*.

ΔH_{rxn}	ΔS_{rxn}	ΔG_{rxn}	Reaction is ...
−	+	−	
+	−	+	
−	−	−	
		+	
+	+	−	
		+	



Free-Energy Changes and Temperature

18.5

At equilibrium:

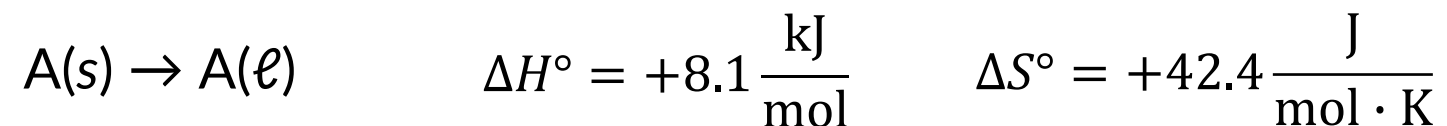
The “low” and “high” temperatures are defined by solving for T .

This is useful for calculating the temperature at which phase transitions occur!

IN-CLASS QUESTION 1

Learning Objective(s): 18.8

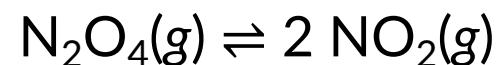
What is the melting point (in K) of an unknown compound A given the standard changes in enthalpy and entropy for the process.



IN-CLASS QUESTION 2

Learning Objective(s): 18.6, 18.7

At a given temperature and pressure, the following reaction is endothermic.

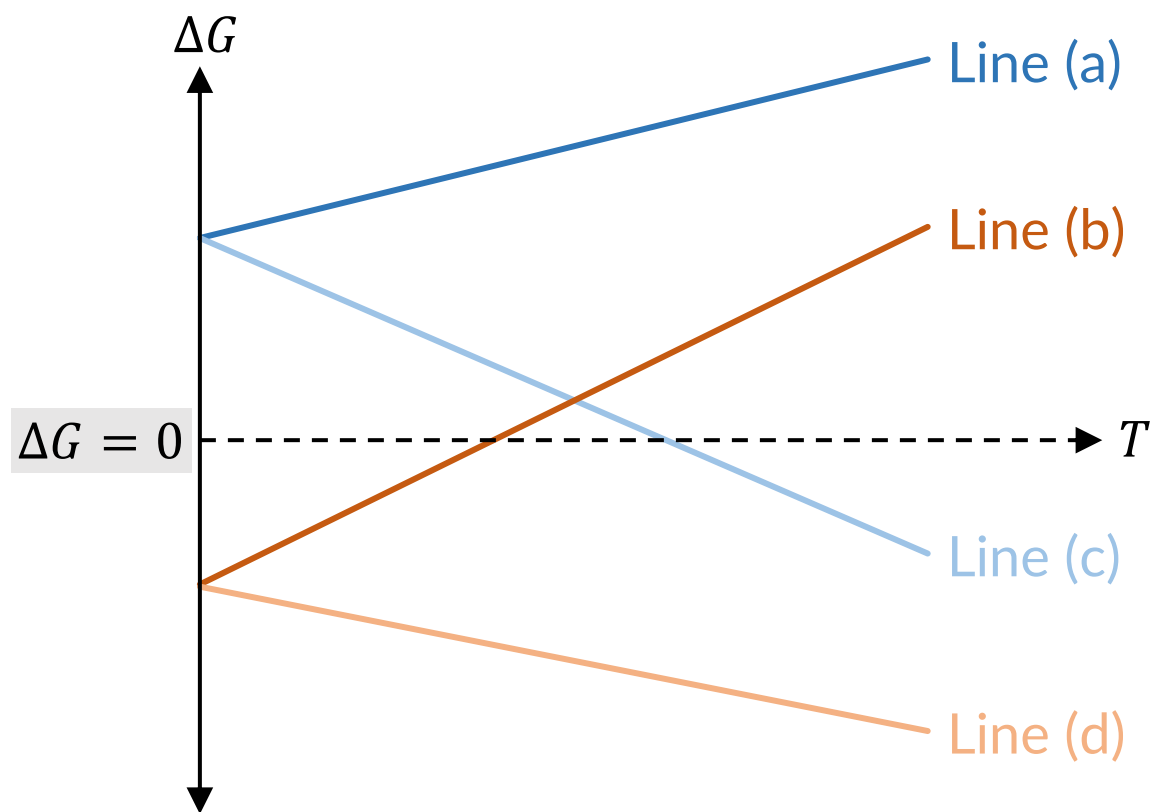


As written, this reaction is ...

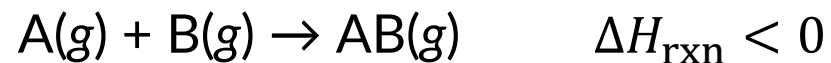
- ☐ Spontaneous at all temperatures.
- ☐ Spontaneous at high temperatures.
- ☐ Spontaneous at low temperatures.
- ☐ Nonspontaneous at all temperatures.

IN-CLASS QUESTION 3

Learning Objective(s): 18.6, 18.7



Which line on the graph of ΔG versus T (given to left) correlates to the following hypothetical thermochemical reaction?



Gibbs Free Energy and Equilibrium

18.6

Under standard conditions: ΔG°

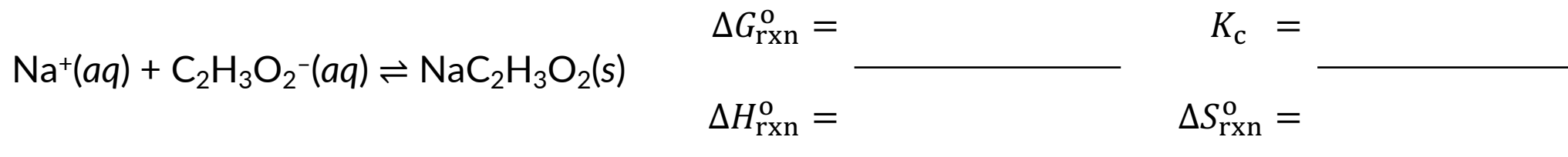
Under non-standard conditions: $\Delta G = \Delta G^\circ + RT \cdot \ln Q$

When a reaction is at equilibrium: $\Delta G = 0$ and $Q = K$

EXAMPLE PROBLEM

Learning Objective(s): 18.7, 18.8, 18.9

Calculate $\Delta G_{\text{rxn}}^{\circ}$ (in kJ/mol) and K_c for the following reaction at 25 °C.



Substance	$\Delta H_f^{\circ} \left(\frac{\text{kJ}}{\text{mol}} \right)$	$S^{\circ} \left(\frac{\text{J}}{\text{mol} \cdot \text{K}} \right)$	$\Delta G_f^{\circ} \left(\frac{\text{kJ}}{\text{mol}} \right)$
$\text{Na}^+(aq)$	-240.3	+58.5	-261.905
$\text{C}_2\text{H}_3\text{O}_2^-(aq)$	-485.10	+104.5	-369.3
$\text{NaC}_2\text{H}_3\text{O}_2(s)$	-709.32	+138.1	-607.7

BONUS QUESTION

Assessed on: Summer 2023, Final Exam

The chemical reaction below is spontaneous at 800 K and nonspontaneous at 400 K.



Which statement is true?

- ☐ The reaction is endothermic.
- ☐ The reaction is exothermic.
- ☐ There is not enough information to determine if this reaction is endothermic or exothermic.